

Alexander Belik

Theoretical Physics | Quantum Thermodynamics | Scientific Computing

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PROFILE

Theoretical physics graduate and current M.Sc. High-Performance Computing student at Trinity College Dublin. I build research platforms, GPU and HPC simulation tooling, machine-learning systems, and full-stack products across theoretical physics, scientific computing, and data-heavy software.

EDUCATION

M.Sc. High-Performance Computing

Trinity College Dublin, current

B.A. Theoretical Physics

Trinity College Dublin, 2022-2026

EXPERIENCE

Co-founder and Engineer

Aug 2026 - Present

Inhabis

- Building a local-first operating layer for connected homes: an embedded Matter controller with one persistent fabric per home, and an Android companion for BLE and Thread onboarding.
- Working on the planning and safety layers that check every action against household policy and verify the result, and on solar-surplus EV charging, solar forecasting and a learned room thermal predictor.

EuroHPC Student Ambassador

Aug 2026 - Sep 2026

EuroHPC Demo Lab 2026

- As a EuroHPC student ambassador, built fourteen GPU physics and AI demos and a walk-up viewer for the Demo Lab, rendered on Leonardo A100 nodes and Brain++ Discoverer GB200 nodes and replayed on a public stand.
- Wrote the SLURM job templates and run bundles that move saved runs between the clusters and a stand laptop.

Scientific Software Developer

May 2026 - Sep 2026

Trinity College Dublin

- Built a digital lab notebook platform storing experimental measurements as structured, auditable records.
- Developed GPU-accelerated physics demonstrations that run headlessly on the Leonardo EuroHPC system and on Brain++ Discoverer GB200 nodes, for the EuroHPC Demo Lab 2026.

Teaching Assistant

Oct 2025 - May 2026

Trinity College Dublin

- Led weekly tutorials and problem-solving sessions aligned with lectures.
- Graded assignments and exams with clear, constructive feedback.

Research Assistant

Jun 2025 - Sep 2025

Trinity College Dublin, CRANN Institute

- Designed Echo State Network reservoir models for quantum time-evolution data.
- Analysed model stability, convergence, and predictive accuracy across hyperparameters.

Software Engineer (EdTech)

Jun 2024 - Dec 2025

Grinds360

- Built Python automation tools for video and interactive lesson workflows.
- Contributed to UX iteration in Figma and full-stack product features.

Experimental / Computational Research Assistant

May 2024 - Sep 2024

Trinity College Dublin

- Created a Python spectroscopy control system for ferrofluids and ferronematic liquid crystals.
- Built automated acquisition and analysis workflows for repeated experimental data.

Exam Corrector

Jul 2021 - Jan 2024

The Dublin Academy of Education

- Corrected exams and coursework and supported marking workflows.

SELECTED WORK

CAPSTONE THESIS

Locally Thermal from Global Athermality

My capstone on quantum states that look thermal from every local view while their correlations still hold non-equilibrium structure.

- Characterized locally thermal states as affine slices of the positive semidefinite cone

Tools: Quantum thermodynamics, Linear algebra, SDP methods, Python, LaTeX

PHYSICS-INFORMED ML

EchoState and Heisenberg Chain

A neural network that learns how a quantum spin chain evolves in time, built on echo state networks in PyTorch.

- General-purpose ESN package with configurable reservoirs, feedback, washout, and mixed precision

Tools: PyTorch, NumPy, SciPy, QuTiP, Optuna, Python

[View on GitHub](#)

HPC AND GPU SIMULATION

EuroHPC Demo Lab

Fourteen physics and AI demos I rendered on Leonardo and Discoverer for the EuroHPC Demo Lab, played back on a public stand.

- Fourteen solvers spanning lattice-Boltzmann flow, particle-mesh cosmology, direct N-body gravity, black-hole lensing, reaction-diffusion, plasma control, molecular dynamics, and two evolving-AI games

Tools: Python, CuPy / CUDA, PyTorch, NumPy, SLURM, JavaScript

[View on GitHub](#)

PUBLISHED EXPERIMENTAL RESEARCH

Ferronematic Liquid-Crystal Research

Instrument control and data analysis for ferronematic liquid-crystal experiments, work that fed into a published paper.

- Python control system driving a spectroscopy instrument through its Windows acquisition software

Tools: Python, Instrumentation, Data acquisition, Matplotlib, Research workflow

RSC publication

TOOLKIT

Scientific computing: Python, NumPy, PyTorch, Mathematica, C / C++, Matplotlib, PySCF, Optuna

HPC and GPU: CUDA / CuPy, SLURM, CMake, Linux, Benchmarking, Parallel pipelines, N-body and lattice-Boltzmann solvers

Web and product: React, TypeScript, JavaScript, Node.js, Vite, Tailwind CSS, FastAPI, three.js, Playwright, Figma, Vercel

Home and mobile: Matter, Thread networking, BLE commissioning, Kotlin, Android, Jetpack Compose, Home energy and thermal models

Data and infrastructure: PostgreSQL, SQLite, Docker, Git / GitHub, GitHub Actions, Data pipelines

AI systems: Ollama, LangGraph, Local LLM workflows, Agent orchestration, Tool routing, Speech to text

Research practice: LaTeX, Quantum mechanics, Statistical mechanics, QFT, Semidefinite programming, Data analysis, Reproducibility